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Plasmon Induced Transparency in Graphene-based Side Coupled two Rectangular Nano-cavities-like structures

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Abstract

We investigate numerically the plasmon analogue of electromagnetic induced transparency (EIT) in a compact graphene-based waveguide coupled to two rectangular nano-cavities-like structures made out of graphene nanoribbons. The cavities are placed symmetrically on each side of the bus waveguide, giving rise to a V-like state in analogy with three level atomic systems. In this configuration, the cavities are directly coupled to the waveguide, and consequently both play the role of the bright mode. By slightly detuning the Fermi energy level of the cavities, we show the possibility of generating a narrow transparency window, that's, the plasmon induced transparency (PIT). The resonance peak occurs as a consequence of destructive interference between the cavities acting like coupled radiative oscillators. We also demonstrate that by appropriately shifting (symmetrically) the Fermi energy levels of the cavities, one could dynamically tune the resonance peak frequency. The proposed structure may help the design of highly integrated optical devices.

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1. Introduction

Electromagnetically induced transparency (EIT) is a quantum effect occurring in atomic systems caused by destructive interference between two excitation pathways to an upper atomic level [1]. The atomic medium previously opaque becomes transparent within a narrow spectral region with low absorption and steep dispersion.

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These appealing properties have plenty potential applications in slowing light, enhancing optical nonlinearity and data storage [2–4].

However, the realization of EIT phenomenon in atomic systems is quite challenging due to the extreme required conditions at the experimental level [5]. To overcome these difficulties, a lot of efforts have been made to mimic the EIT-effect in classical optics. In particular, during the last decade a great attention has been paid to the use of Metal-Insulator-Metal (MIM) based materials to achieve the plasmon induced transparency (PIT), which is simply the plasmonic analogue of EIT. Usage of MIM structures is due to their unique ability of light confinement at the nanoscale domain. Graphene on the other hand is one carbon atom thick material in a honeycomb lattice with outstanding electrical and optical properties. Specifically, one of the key advantages of conceiving graphene-based devices is the fact that its electrical conductivity can be dynamically tuned by changing the Fermi energy level [6], which makes it ideal for realizing tunable optical devices.

Among many plasmonic or graphene-based plasmonic structures that have been proposed lately to realize EIT-like resonance, one can cite: planar metamaterials [7, 8], microbubble whispering-gallery mode resonators [9], micro- and radio wave circuits [10–12], hybrid metal-graphene metamaterial [13], graphene-based on-chip plasmonic nanostructure [14], high-quality factor tunable-EIT plasmonic system [15], liquid crystal metamaterial-based modulator at the terahertz range [16] and tunable graphene-based nano-device for sensing, power splitting and demultiplexing at the mid-infrared [17]. Within the spirit of the above studies, we propose in this work an ultra-compact graphene-based nanoscale waveguide for the realization of dynamically tunable PIT-like transmission in the THz regime.

The structure consists simply in a graphene waveguide coupled to two rectangular nanocavities-like structures made out of two identical graphene nanoribbons each, and inserted symmetrically on each side of the main waveguide, giving rise to a V-like state in analogy with three level atomic systems [1]. In this configuration, both cavities acting like radiative oscillators play the role of bright modes.

2. Numerical method of simulation

Figure 1 presents a sketch of the proposed geometry design. The cavities are characterized respectively by their lengths and widths, L_1 and w_1 for the upper cavity, and L_2 and w_2 for the lower one. They are far apart from the bus waveguide by the same distance, $d_1 = d_2 = 185 \text{ nm}$, which is kept unchanged throughout this study. In order to suit practical applications, the graphene nanostructures are embedded in a dielectric material whose electric permittivity is $\epsilon = 2.25$.

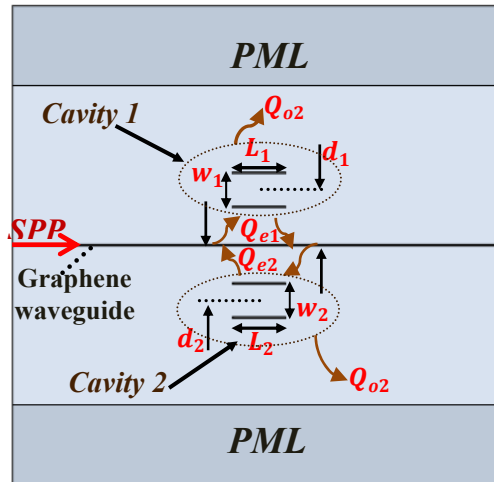


Fig. 1. Schematic diagram of a graphene-based waveguide structure coupled to two graphene rectangular nanocavities-like structures, placed symmetrically on each side of the bus waveguide.

On the other hand, graphene in our model is treated as a very thin anisotropic material with a thickness $d = 1 \text{ nm}$. Its dielectric constant tensor is such that $\epsilon_{\perp} = 2.5$ is the out-of-plane permittivity while $\epsilon_{//} = 1 + \sigma / \epsilon_0 \omega d$ is the in-plane permittivity, where σ is the surface conductivity of graphene, ϵ_0 is the vacuum permittivity and ω is the light angular frequency. In the terahertz frequency domain σ takes the form of a Drude-like expression, that is, $\sigma = ie^2 E_F / \pi \hbar^2 (\omega + i\tau^{-1})$ [18, 19], where e , E_F , $\hbar$, τ are the electron charge, the Fermi energy level, the reduced Planck constant and the relaxation time, respectively. The relaxation time is related to the Fermi velocity, $v_F = 10^6 \text{ m/s}$, and to the DC mobility μ as $\tau = \mu E_F / e v_F^2$. We assumed a value of $10^5 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$ for μ as in [18] for all graphene nanostructures considered here. Numerical simulations are performed using the finite element method (FEM) via Comsol Multiphysics Package. In order to avoid any reflection perturbing the system, we did use PML layers on the bottom and upper sides of the structure (as depicted in Fig. 1). Only transverse magnetic (TM) SPP modes are to be considered throughout this paper. The Fermi energy levels of the cavities are defined as E_{F1} for the upper cavity, and E_{F2} for the lower one. A fixed value is chosen for the bus waveguide such as $E_{F0} = 140 \text{ meV}$. An eigen SPP-mode of the graphene waveguide is excited and launched from the left input port of the structure, and the transmission spectrum is then recorded at the output port at the right side.

3. Results and Discussion

Figures 2(a) and 2(b) give the transmission spectra of the structure depicted in Fig. 1 in the case where either the upper cavity or the lower one is present alone within the geometry, respectively. The values of the geometrical parameters in these simulations are such as: $L_1 = L_2 = 250 \text{ nm}$ and $w_1 = w_2 = 110 \text{ nm}$. As concerning the Fermi energy levels of the cavities we took, $E_{F1} = 140 \text{ meV}$ (upper cavity) and $E_{F2} = 170 \text{ meV}$ (lower cavity). A fitting curve based on temporal coupled mode (CMT) theory is also shown in Figs. 2(a) (red circles) and 2(b) (green circles); despite some discrepancies one can notice a fairly good agreement between the numerical and theoretical data. The CMT results are obtained as follows: if one denotes by ω_i the resonance frequency of the cavity i ($i = 1, 2$), then the complex transmission and reflection coefficients are given by [20]:

$$t_i = j(\omega_i - \omega) + \omega_i / 2Q_{oi} / (j(\omega_i - \omega) + \omega_i / 2Q_{oi} + \omega_i / 2Q_{ei}) \quad (1)$$

$$r_i = -(\omega_i / 2Q_{ei}) / (j(\omega_i - \omega) + \omega_i / 2Q_{oi} + \omega_i / 2Q_{ei}) \quad (2)$$

Here, Q_{oi} and Q_{ei} are the quality factors of cavity i , related to the intrinsic losses and to the waveguide coupling strength magnitude respectively. The theoretical transmission spectrum (of cavity i) is given by $T_i = |t_i|^2 \times |t_0|^2$, where $|t_0|^2$ is the transmission spectrum of the bus waveguide without any cavity. The multiplication by the latter enables the catching of the asymmetric profile of the main waveguide transmission spectrum as indicated in [20]. Actually, Q_{ei} is related to the quality factor associated with the transmission zero of the resonance by $Q_{ti} = Q_{ei} \times Q_{oi} / (Q_{ei} + Q_{oi})$ [22], where $Q_{ti} = \omega_i / \Delta\omega_i$; $\Delta\omega_i$ being the full width at half maximum (FWHM). As stated in [23], the observed small differences (in terms of transmission values) between theoretical and numerical curves could be attributed to the rough approximation of Q_{ti} and Q_{ei} . However, both theoretical and numerical curves produce a similar profile, and subsequently the provided theoretical fitting can significantly help understanding the transmission spectra obtained numerically.

Now, we focus on the case when both cavities are present in the system (as shown in Fig. 1). The corresponding transmission spectrum is given in Fig. 3(a) for the same values of the geometrical and physical parameters as previously, namely: $L_1 = L_2 = 250 \text{ nm}$, $w_1 = w_2 = 110 \text{ nm}$ and $E_{F1} = 140 \text{ meV}$ (for the upper cavity) and $E_{F2} = 170 \text{ meV}$ (for the lower cavity). As a matter of comparison, the transmission spectra of each cavity resonator alone (black and red curves) are also plotted in Fig. 3(a). One notes the emergence of a resonance peak at the frequency $f_r = 8.53 \text{ THz}$ squeezed between two dips at, $f_1 = 8.2 \text{ THz}$ and $f_2 = 9.1 \text{ THz}$. These dips originate from the cavities resonances, while the PIT-like transmission at f_r arises as a consequence of destructive interference between the two cavities acting like coupled radiative oscillators. Indeed, in Fig. 3(d) is shown the z -component of the magnetic field (H_z) map within the structure at the PIT-frequency, and it can be clearly seen that both cavities

are excited and oscillate in anti-phase in the vicinity of the main waveguide, which results in an unperturbed transmission.

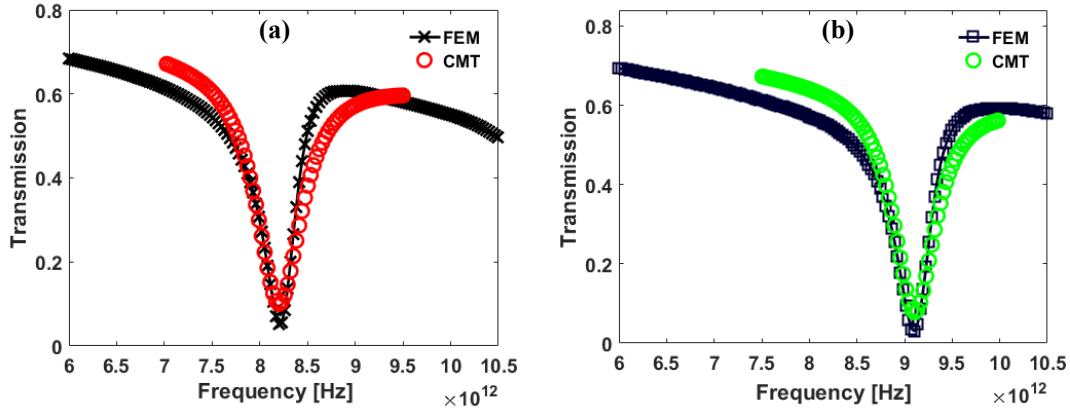
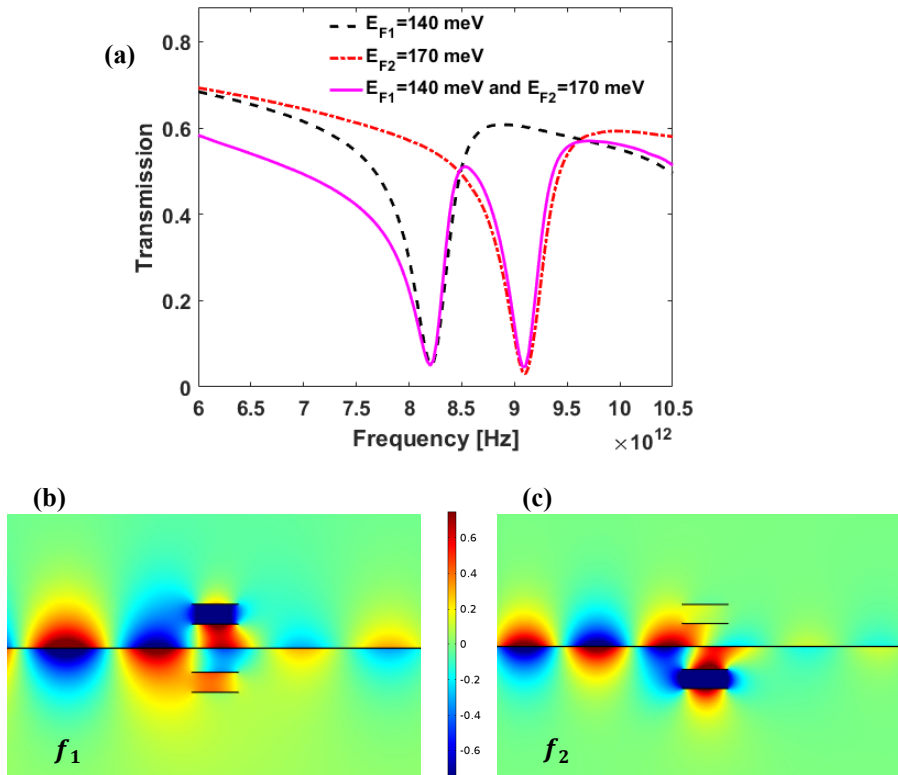


Fig.2. (a) Transmission spectrum of the structure depicted in Fig. 1 when only the upper cavity is present, for $L_1=250$ nm, $w_1=110$ nm and $E_{F1}=140$ meV. The black solid lines with black stars refer to numerical results whereas the red circles represent the theoretical fitting curve by coupled mode theory (CMT). (b) Same as in (a) but for the lower cavity such that: $L_2=250$ nm, $w_2=110$ nm and $E_{F2}=170$ meV. The Solid dark blue line with open squares represents the numerical curve, while the green open circles refer to the fitting curve.

Figures 3(b) and 3(c) present the H_z field distribution at the transmission dips frequencies f_1 and f_2 . At these frequencies, the incident waves are trapped in each cavity and do not propagate in the system. This effect has already been reported in [21].



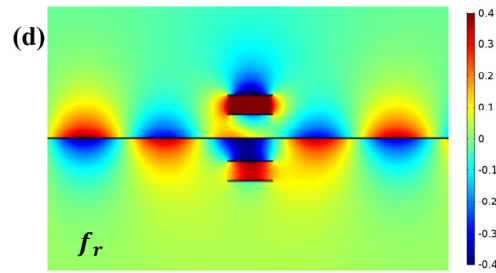


Fig.3. (a) Transmission spectra of the structure of Fig. 1, with $L_1 = L_2 = 250$ nm, $w_1 = w_2 = 110$ nm and $E_{F1} = 140$ meV and $E_{F2} = 170$ meV (pink solid line). Black dashed line refers to the transmission spectrum of a single cavity placed in the upper side of the waveguide, with $E_{F1} = 140$ meV. Similarly, the dotted-dashed red line correspond to the transmission spectrum when a single cavity is in the lower side, such that $E_{F2} = 170$ meV. (b), (c) H_z -field map distribution within the structure at the transmission dips (see the pink solid curve of (a)) frequencies, $f_1 = 8.2$ THz and $f_2 = 9.1$ THz. (d) Same as in (c) and (d) but at the PIT-resonance frequency, $f_r = 8.53$ THz. Notice the different colour scales in (d) as compared to (b) and (c).

In order to catch the main features of the PIT-resonance peak, a theoretical model based on coupled mode theory is carried-out. The theoretical transmission spectrum of the system is given by [20]:

$$T = |t_0|^2 \times |t_1 t_2|^2 / |1 - r_1 r_2|^2 \quad (3)$$

where, t_i and r_i ($i = 1, 2$) are the complex transmission and reflection coefficients of cavity i (provided above). For the same reasons as stated above, the transmittance of the system is multiplied by the waveguide transmission (with no cavity), $|t_0|$. Figure 4 gives the numerical transmission (black solid curve with diamonds) spectrum of the system along with the fitting curve (green open circles) based on the CMT model where the all the geometrical and physical parameters have the same values given in the previous paragraph. The theoretical curve captures the essential features of the numerical PIT (transmission) curve, with some discrepancies beyond the positions of resonances and anti-resonances. As explained above, this may be related to the rough estimation of quality factors Q_{li} and Q_{ei} , which are even more roughly determined when both cavities are coupled to the bus waveguide. Indeed, in this instance, $Q_{ti} = \omega_i / \Delta\omega_i$ should be evaluated from the FWHM of the transmission dip associated with cavity i in the transmission spectrum. However, this quantity is not very well defined in this case, so does the value of Q_{ti} , and subsequently Q_{ei} .

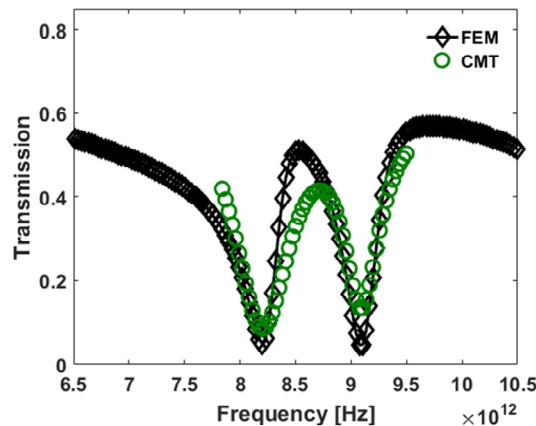


Fig. 4. Comparison of the transmission spectrum of the structure sketched in Fig. 1 (with the same geometrical and physical parameters values indicated in the text) obtained numerically and the theoretical one based on the CMT-model.

Now, we would like to show the possibility of dynamically tuning the PIT-resonance frequency of the system. In practice, this is realized by applying a gate voltage on the cavities. For that matter, we calculated the transmission spectra for different values of E_{F1} and E_{F2} . This could be done in two different manners: *i)* by shifting simultaneously the values of E_{F1} and E_{F2} from a given Fermi energy levels, *ii)* or by changing the value of one of them and keeping the other one unchanged, which controls the resonance width (or equivalently the quality factor of the transmission peak). In Fig. 5, we present the transmission spectra for a simultaneous shifting of the Fermi energy levels of the cavities, namely for: ($E_{F1} = 122 \text{ meV}$, $E_{F2} = 152 \text{ meV}$), ($E_{F1} = 140 \text{ meV}$, $E_{F2} = 170 \text{ meV}$) and ($E_{F1} = 158 \text{ meV}$, $E_{F2} = 188 \text{ meV}$). We observe an almost linear and important blue shift of the PIT-frequency resonance with the increase of the Fermi energy levels. This can be understood knowing that the frequency resonance of a localized SPP mode of a graphene ribbon-like material is given by $\propto (e/2\pi\hbar c) \times \sqrt{E_F/L}$ [18], where L and E_F are the ribbon width and Fermi energy level respectively. On the other hand, the studied cavities are actually made of graphene ribbons-like materials connected in a rectangular shape. As a result, increase E_F will augment the resonance frequencies of the dips and, so does the PIT-peak frequency squeezed between the dips.

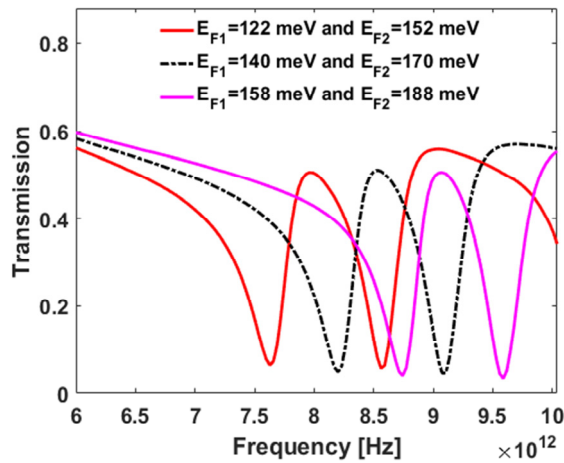


Fig. 5. Transmission spectra of the system for three different values of E_{F1} and E_{F2} such as: ($E_{F1} = 122 \text{ meV}$, $E_{F2} = 152 \text{ meV}$), ($E_{F1} = 140 \text{ meV}$, $E_{F2} = 170 \text{ meV}$) and ($E_{F1} = 158 \text{ meV}$, $E_{F2} = 188 \text{ meV}$).

It is well established that within the transparency window associated with an EIT-resonance, a steep dispersion takes place, giving rise to slow light phenomenon inside this frequency range. To study this effect, we plotted in Fig. 6(a), the transmission phase versus the frequency for $L_1 = L_2 = 250 \text{ nm}$, $w_1 = w_2 = 110 \text{ nm}$, $E_{F1} = 140 \text{ meV}$ and $E_{F2} = 170 \text{ meV}$. As predicted, the phase drops twice around the transmission dips frequencies. However, the phase drops do not reach π due to the large absorption in the graphene. In Fig. 6(b), is given the delay time $\tau = -\frac{d\phi}{d\omega}$, which clearly indicates the phase drops signature around the dips frequencies (where τ turns negative) and a certain delay time of the signal within the PIT window as τ increases in this frequency range. Let us point out that τ represents the time of capture of the light waves inside the cavities. Also, in Fig. 6(c) we present the relative group velocity defined as V_g/c , where V_g is the group velocity given by $V_g = L/\tau$. L ($= 2.325 \mu\text{m}$) is the (effective) distance travelled by the signal, which is approximately equal to the total length of the system, and c is the speed of light in vacuum. Figure 6(c) shows a similar trend as in [24], where the relative group velocity turns negative around the dips frequencies and as expected falls down (subliminal effect) by a non-negligible amount within the transparency window. This can be seen in Fig. 6(d), where a zoom in is presented around the frequency domain associated with the PIT resonance.

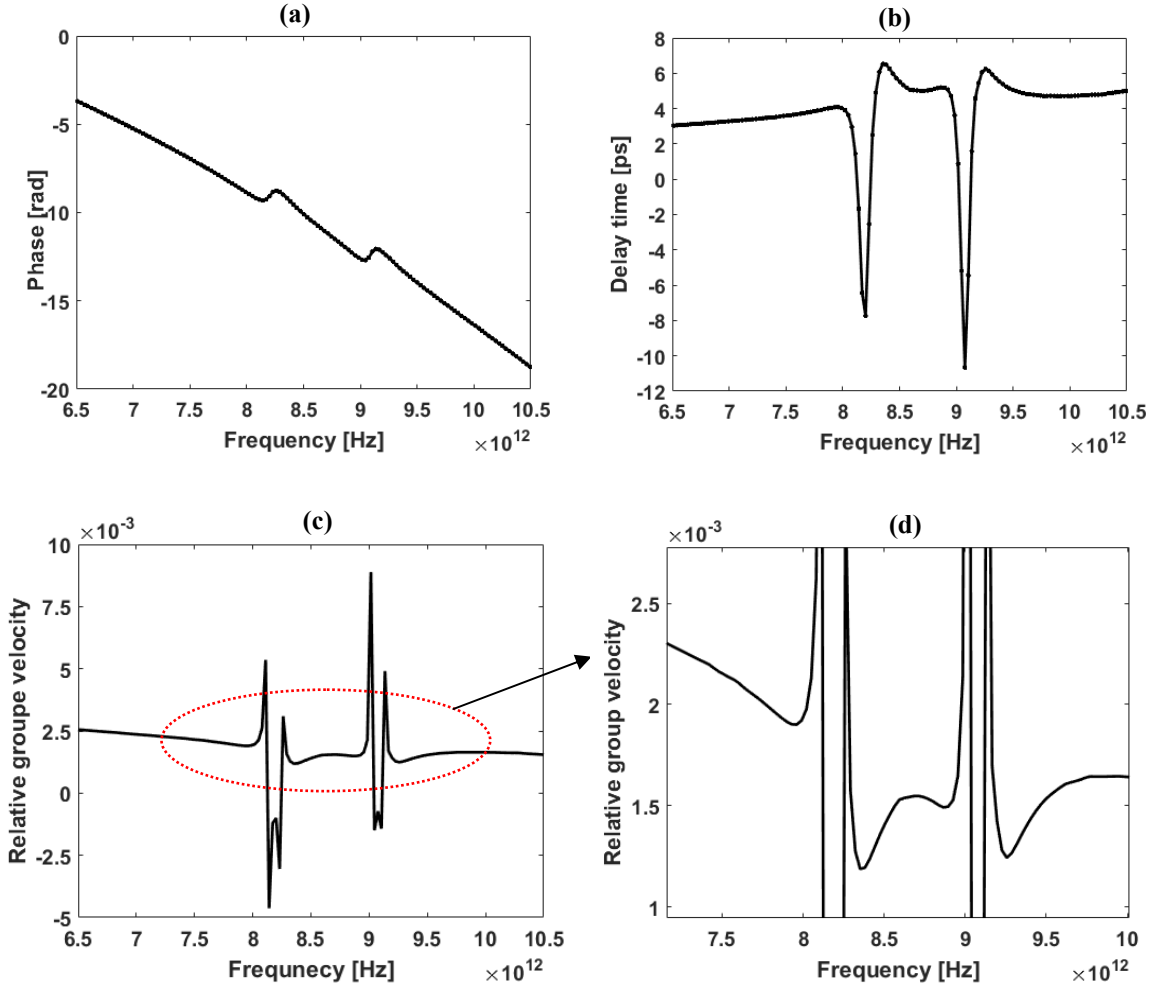


Fig. 6: (a) Transmission phase of the structure depicted in Fig. 1 versus the frequency with $L_1 = L_2 = 250 \text{ nm}$, $w_1 = w_2 = 110 \text{ nm}$, $E_{F1} = 140 \text{ meV}$ and $E_{F2} = 170 \text{ meV}$. (b), (c) Same as in (a) but for the time delay and the relative group velocity V_g/c (c being the velocity of light in vacuum). (d) Zoom in around the PIT resonance frequency range in Fig. 6(c).

4. Conclusion

In this paper, we proposed an ultra-compact graphene-based waveguide structure for the realization of plasmon induced transparency (PIT) in a V-type like configuration (in analogy with three level atomic systems). The structure is composed of a bus waveguide coupled to two rectangular nano-cavities made out of graphene, and put symmetrically aside. The cavities and the waveguide are embedded in a dielectric material. We showed that by slightly detuning the Fermi energy levels of the cavities, a narrow transmission window is generated, that's the PIT-resonance, which is the plasmonic analogue of EIT. The origin of the resonance is explained as being caused by destructive interference of the coupled cavities (oscillating in anti-phase), acting like radiative oscillators. A theoretical model based on coupled mode theory (CMT) is carried out for fitting of the numerical data. Overall, the CMT-results are in a good agreement with the simulated data. We demonstrated the possibility of dynamically tuning the PIT-resonance frequency by shifting the Fermi energy levels of the cavities. The dispersion effects accompanying the PIT-transmission (within its frequency range) are also shown. The proposed structure may help the design of highly integrated optical devices such as: nanoscale optical filters, nano-sensors and high-speed switches.

References

- [1] M. Fleischhauer, A. Imamoglu and J.P. Marangos, *Rev. Mod. Phys.* 77 (2005) 633-41.
- [2] J. Zhang, G. Hernandez and Y. Zhu, *Opt. Lett.* 33 (2008) 46-48.
- [3] G. Heinze, C. Hubrich and T. Halfmann, *Phys. Rev. Lett.* 111 (2013) 033601-5.
- [4] M.M. Alotaibi, Hessa and B.C. Sanders, *Phys. Rev. A* 94 (2016) 053832-11.
- [5] N. Papasimakis, V.A. Fedotov, N.I. Zheludev and S.L. Prosvirnin, *Phys. Rev. Lett.* 101 (2008) 253903-4.
- [6] S.X. Xia, X. Zhai, L.L. Wang, B. Sun, J.Q. Liu and S.C. Wen, *Opt. Express* 24 (2016) 17886-14.
- [7] J. Wu, B. Jin, J. Wan, L. Liang, Y. Zhang, T. Jia, C. Cao, L. Kang, W. Xu, J. Chen, P. Wu, *Appl. Phys. Lett.* 99 (2011) 161113-3.
- [8] X. Zhang, N. Xu, K. Qu, Z. Tian, R. Singh, J. Han, G.S. Agarwal and W. Zhang, *Sci. Rep.* 5 (2015) 10737-9.
- [9] Z. Chen, R. Hu, L. Cui, L. Yu, L. Wang and J. Xiao, *Opt. Commun.* 320 (2014) 6-11.
- [10] Y. Yang, S. Saurabh, J. Ward, S.N. Chormaic, *Opt. Lett.* 40 (2015) 1834-7.
- [11] A. Mouadili, E.H. El Boudouti, A. Soltani, A. Talbi, A. Akjouj, B. Djafari-Rouhani, *J. Appl. Phys.* 113 (2013) 164101-11.
- [12] A. Mouadili, E.H. El Boudouti, A. Soltani, A. Talbi, A. Akjouj, B. Djafari-Rouhani, A. Akjouj, K. Haddadi, *J. Phys.: Condens. Matter* 26 (2014) 505901-12.
- [13] X. Yan, T. Wang, S. Xiao, T. Liu, H. Hou, L. Cheng and X. Jiang, *Sci. Rep.* 7 (2017) 13917-10.
- [14] P. Qiu, W. Qiu, Z. Lin, H. Chen, J. Ren, J.X. Wang, Q. Kan and J.Q. Pan, *Nanoscale Res. Lett.* 12 (2017) 374-8.
- [15] H. Lu, X. Gan, D. Mao, B. Jia, J. Zhao, *Sci. Rep.* 8 (2018) 1558-9.
- [16] J. Wang, H. Tian, Y. Wang, X. Li, Y. Cao, L. Li, J. Liu and Z. Zhou, *Opt. Express* 26 (2018) 5769-8.
- [17] S. Asgari, Z.G. Kashani and N. Granpayeh, *J. Opt.* 20 (2018) 045001-11.
- [18] L. Wang, W. Li and X. Jiang, *Opt. Express* 40 (2015) 2325-4.
- [19] F.H.L. Koppens, D.E. Chang and F.J.G. De Abajo, *Nano Lett.* 11 (2011) 3370-8.
- [20] Q. Lin, X. Zhai, L. Wang, B. Wang, G. Liu and S. Xia, *EPL* 111 (2015) 34004-6.
- [21] A. Noul, Y. Pennec, A. Akjouj, B. Djafari-Rouhani and L. Dobrzynski, *J. Phys.: Condens. Matter* 21 (2009) 375301-6.
- [22] H. Lu, X. Liu, D. Mao, *Phys. Rev. A* 85 (2012) 053803-7.
- [23] S. Zhan, D. Kong, G. Cao, Z. He, Y. Wang, G. Xu, H. Li, *Solid State Commun.* 174 (2013) 50-5.
- [24] M.P. Hokmabadi, E. Philip, E. Rivera, P. Kung and S.M. Kim, *Sci. Rep.* 5 (2015) 15735-11.